Aritra Bhattacharya

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Research Summary

My research centers on **algebraic combinatorics**, with particular emphasis on **symmetric functions**, Macdonald polynomials and their specializations (Hall-Littlewood, q-Whittaker, LLT), and **Kazhdan–Lusztig theory**. In my research I have extensively used combinatorial models involving Young tableaux and Dyck paths, as well as algebraic tools from finite, affine, and double affine Hecke algebras, and incorporated geometric techniques from perverse sheaves and mixed Hodge modules on flag varieties. I regularly use `SageMath` for computational experimentation to formulate and test conjectures.

Education and Employment

2026–present	Postdoctoral Fellow, Beijing Institute of Technology, China. *Mentor: David Wang*
2024–2026	Postdoctoral Fellow, Beijing International Center for Mathematical Research, Peking University, China. *Mentor: Yibo Gao*
2018–2024	Ph.D. in Mathematics, The Institute of Mathematical Sciences, Chennai, India. *Advisor: Sankaran Viswanath*
2016–2018	M.Sc. in Mathematics, Savitribai Phule Pune University, Pune, India.
2013–2016	B.Sc. with Honors in Mathematics, University of Calcutta, Kolkata, India.

Publications

1. **q-Whittaker polynomials: bases, branching and direct limits**, with T. V. Ratheesh and Sankaran Viswanath,
   European Journal of Combinatorics, Vol. 138, December 2026
   arXiv: 2412.00116, 2024.

2. **A tableaux formula for q-rook numbers**, with Tirtharaj Basu,
   Algebraic Combinatorics, Vol. 9 (2026) No. 1,
   arXiv: 2507.00766, 2025.

3. **Equating Inv-Quinv formulas for the q-Whittaker and modified Hall-Littlewood functions**,
   The Electronic Journal of Combinatorics, Volume 32, Issue 4 (2025),
   arXiv: 2412.09929, 2024.

4. **Monomial expansions for q-Whittaker and modified Hall-Littlewood polynomials**, with T. V. Ratheesh and Sankaran Viswanath,

FPSAC 2024 extended abstract, Séminaire Lotharingien de Combinatoire, 91B.61 (2024), arXiv: 2311.07904, 2023.

5. **Clebsch–Gordan coefficients for Macdonald polynomials**, with Arun Ram, Algebras and Representation Theory, Volume 27, 2024, arXiv: 2310.10846, 2023.

6. **Haglund's positivity conjecture for multiplicity one pairs**, Séminaire Lotharingien de Combinatoire, vol. 87, 2023, arXiv: 2205.11802, 2022.

Preprints

1. **Changing parameters in Hall-Littlewood functions**, with Ziyi Dai, Qilin Hou and Edward Tong, arxiv: 2610.41155, 2026.

2. **On factorization of matrix of Kazhdan–Lusztig polynomials**, with A. Mishra and S. Srivastava, arXiv: 2602.19508, 2026.

3. **The monomial expansion formula for Hall-Littlewood P-polynomials**, arXiv: 2407.14652, 2024.

Teaching and Mentoring

- Mentor, PACE 2026 (PKU Algebraic Combinatorics Experience).
- **Lecture series:** *Introduction to Macdonald Polynomials*, Shandong University, October 13–16, 2025.
- **Lecture series:** *Macdonald Polynomials*, The Institute of Mathematical Sciences, September–November 2022.

Presentations

- Talk on *On factorization of matrix of Kazhdan–Lusztig polynomials* at Workshop on Combinatorics, Algebra and Geometry Seminar of Tianyuan Mathematics Research Center, May 13, 2026.
- Talk on *On factorization of matrix of Kazhdan–Lusztig polynomials* at Chinese Academy of Sciences, April 23, 2026.
- Talk on *On factorization of matrix of Kazhdan–Lusztig polynomials* at CART seminar, April 17, 2026.
- Talk on *On factorization of matrix of Kazhdan–Lusztig polynomials* at BICMR, March 9, 2026.
- Talk on *An Introduction to q-Whittaker Polynomials* at BICMR, March 10, 2025.
- Talk on *The Clebsch–Gordan coefficients for Macdonald polynomials* in Algebraic and Combinatorial Methods in Representation Theory, ICTS, November 14, 2023.
- Talk on *The Clebsch–Gordan rules for Macdonald polynomials* at IMSc, October 26, 2023.
- Talk on *Haglund's positivity conjecture for Macdonald polynomials* in Meru Combinatorics Conference, May 29, 2023.

- Talk on *Haglund's positivity conjecture for Macdonald polynomials* in University of Melbourne, March 2023.
- Talk on *Macdonald Symmetric Functions and Haglund's conjecture* in Graduate Online Combinatorics Colloquium, 2022, video: Recorded Talk.
- Talk on *Haglund's Conjecture for Multiplicity one pairs* in Summer School in Algebraic Combinatorics, Kraków, 2022.

Posters

- *The Monomial Expansion formula for Hall-Littlewood Polynomials*, MERU 2024.
- *The Clebsch–Gordan rules for Macdonald polynomials*, ALCOVE 2023.

Service and Organization

- Co-organizer, Weekly IMSc Algebraic Combinatorics Seminar, 2023–2024 (with Amritanshu Prasad).
- Moderator, BIRS-CMI workshop "25w5354: Quivers in representation theory", November 30 – December 5, 2025.